



Design Ecosystems

Overloaded Laundry System

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Table of Contents

Introduction & Problem Context 04

Evidence of the Problem 06

Stakeholder Mapping 08

Systems Cluster Overview 10

System Mapping & Problem Visualisation 22

Influence Map: Effects & Causes 24

Sites of Intervention 26

Leverage Points 27

Gigamap 28

Proposed Interventions 32

Shortlisted Solutions 36

Implementation & Testing 37

Stakeholder Interaction & Feedback 38

RFID Tag Design 40

Addressing Common Concerns 41

Team Members 43

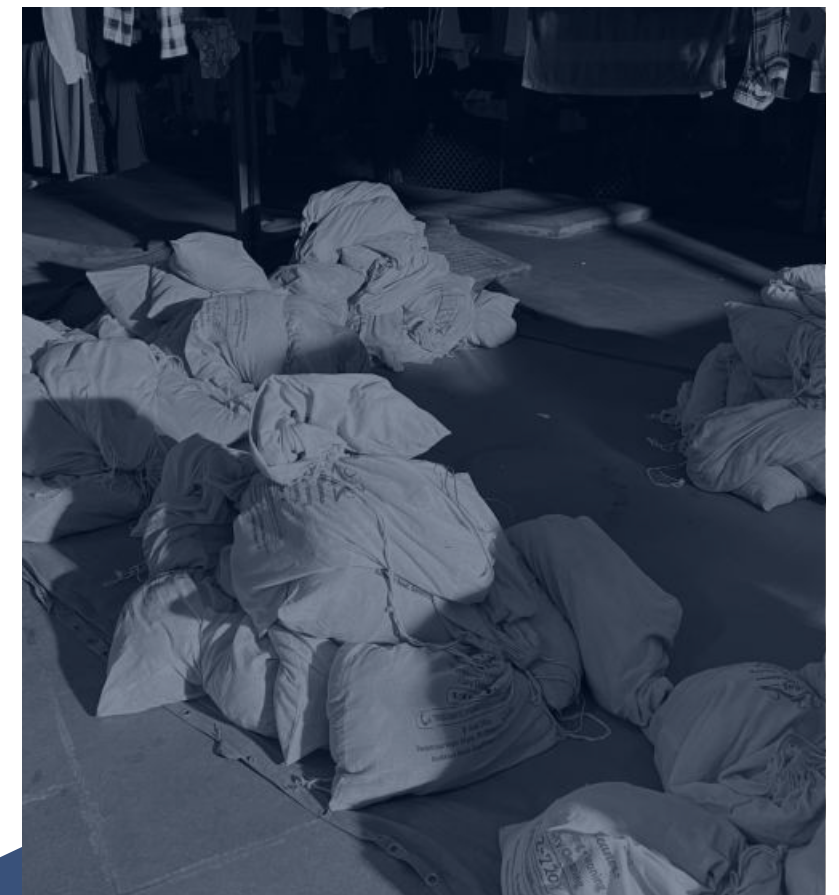
Fixing the Broken Laundry System: A Systemic Intervention Approach

Background

The laundry system in the college hostel is an essential part of student life. However, over the years, this system has become increasingly inefficient and strained, resulting in serious breakdowns in service quality and satisfaction. Key recurring issues include:

- Loss and damage of garments
- Delayed deliveries and unpredictable turnaround times
- Manual and error-prone processes for tagging and logging
- Overburdened staff due to inconsistent manpower allocation
- Lack of student awareness, accountability, and compliance
- Absence of a clear communication or grievance redressal system
- Inadequate infrastructure leading to weather-based disruptions

This has led to a general sense of mistrust, inefficiency, and dissatisfaction among students and staff alike.



Central Variable



Overloaded
Laundry System

Objective

The objective of this study is to approach the laundry system problem through systemic thinking like identifying patterns, variables, feedback loops, and leverage points to propose sustainable interventions that restore accountability, efficiency, and fairness to the entire ecosystem.

Research Question

What factors contribute to the inefficiency and overload in the college laundry system, and how can process improvements create a fairer and more efficient experience for both students and staff?



Overloaded Laundry System – This became the primary variable, functioning as the root node in a complex mesh of interdependent subsystems, eventually forming the core of our system map.

Evidence of the Problem

01



Fill the laundry bags

02



Drop them off

03



Stacking of clothes
at one place

04



The staff then sits sorts and
tags the clothes

13



Students make an entry on
register confirming the
handover with nothing
missing

12



Student check their clothes
to ensure nothing is missing

11



Student collect
their clothes from
the rack

05



The clothes are sorted by color
(light and dark)

06



The clothes are washed

07



They are then dried off

08



Once dried, staff sort and bag
clothes by their tags

10



Bags are kept on racks
designated to hostels

09



Staff irons each cloth, then
stacks and ties it with bag for
delivery

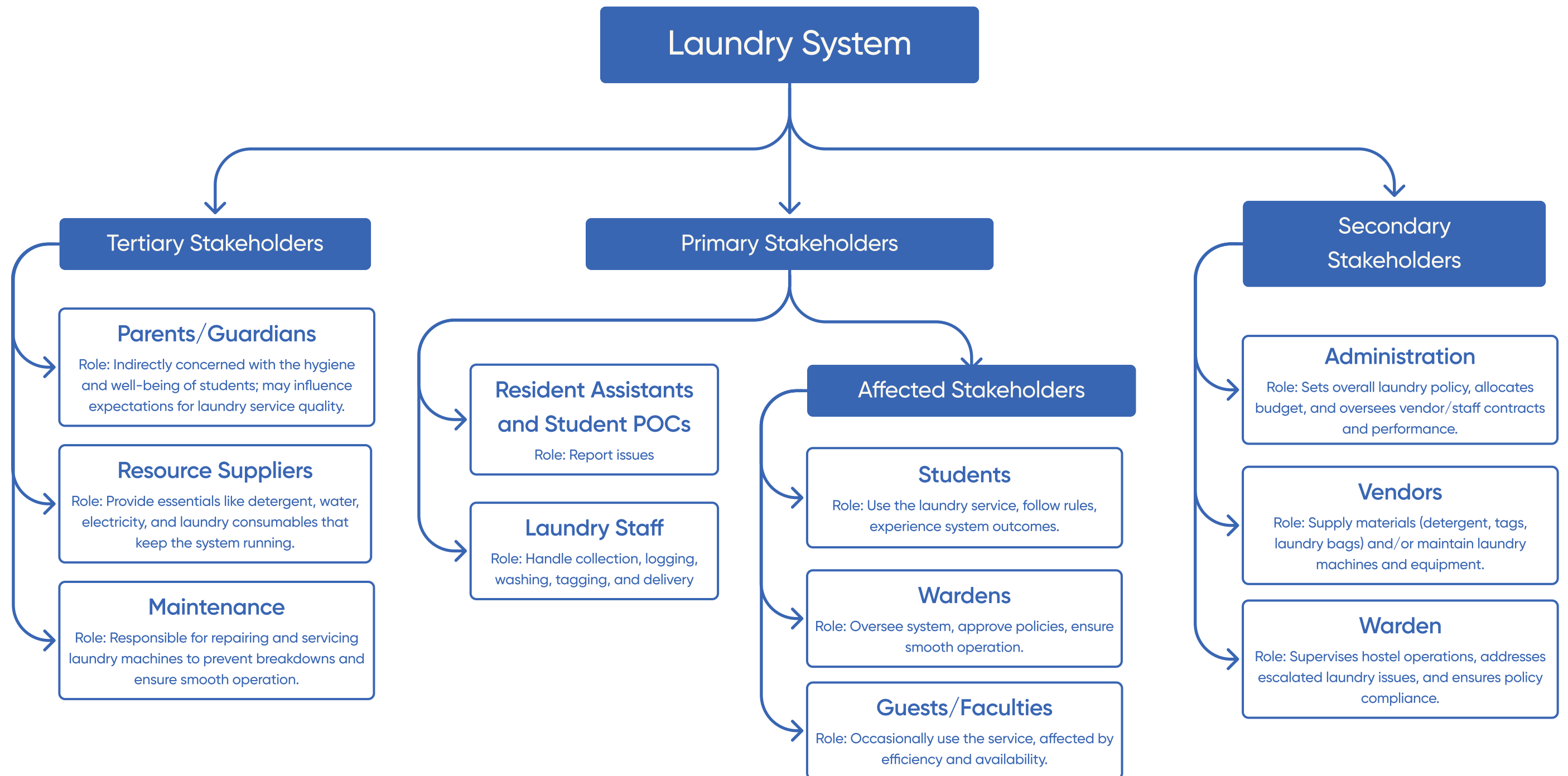
Stakeholder Mapping and Roles

Understanding the System through Stakeholders

To redesign a complex system like the hostel laundry process, it was essential to understand not just the flow of clothes but also the people involved at every step. Stakeholders are individuals or groups that affect or are affected by the system – directly or indirectly. They hold knowledge, face consequences, or have decision-making power that shapes the system's outcomes.

We conducted interviews, observations, and informal conversations to understand the perspectives, pain points, responsibilities, and expectations of each stakeholder. Based on their level of involvement and influence, we classified them into Primary, Secondary, and Tertiary stakeholders.





Systems Clusters Overview

What are clusters?

Clusters are categories of interconnected variables in the system. Grouping them helped identify specific intervention sites and understand the interrelations within the larger system.

Five Main Clusters

Each cluster contains multiple variables that either exacerbate or relieve pressure on the system.

01 Student Behavior & Compliance

Focuses on student actions, accountability, and adherence to laundry-related rules and limitations.

02 Communication & Feedback

Encompasses clarity, transparency, and the availability of reporting or feedback mechanisms between stakeholders.

03 System Design & Infrastructure

Addresses the physical and digital infrastructure that supports or hinders the laundry system's operations.

04 Laundry Process & Operations

Examines the internal workflows, manual tasks, and staff logistics involved in day-to-day laundry handling.

05 System Performance Metrics

Captures quantitative indicators such as delays, complaints, satisfaction, and overall system efficiency.

Student Behavior & Compliance

About the Cluster

This cluster focuses on the habits, awareness, and behavioural patterns of students using the laundry system. Since students are the primary users, their actions directly influence how well the system functions. From the number of clothes submitted to how they perceive and engage with rules, this cluster reveals the critical human dimension of the problem. A lack of compliance or awareness amplifies the system's inefficiencies, while thoughtful behaviour has the potential to significantly ease the burden.



Clothes per Student

The average number of clothes each student submits during a laundry cycle. Higher numbers lead to system overload, especially when the submission limit is not enforced.

Student Accountability

The extent to which individual students take responsibility for their laundry submissions. This includes following rules, submitting on time, and checking for garment return.

Compliance Incentives

Whether the system offers any positive reinforcement or motivation for students who consistently follow the rules. The absence of incentives can lead to indifference or non-compliance.

Rule Violation Rate

The frequency with which students ignore or bypass system rules, particularly the 8-clothes limit. A high violation rate strains the system and undermines fairness.

Perceived Fairness

The overall level of satisfaction students feel with the laundry system, based on timely returns, garment safety, and communication transparency.

Proxy Submissions

The practice of submitting clothes on behalf of friends or roommates, which often leads to breaches in submission limits and loss of accountability.

Awareness of Clothes Limit

The degree to which students know and understand the existing rule that limits them to submitting eight clothes at a time. Many students are unaware or unclear about this guideline.

Student Satisfaction

The overall level of satisfaction students feel with the laundry system, based on timely returns, garment safety, and communication transparency.

Trust in the System

The level of confidence students have in the laundry process that their clothes will be handled properly, returned on time, and not lost or damaged. Low trust can cause students to look for workarounds, adding to system inefficiencies.

Laundry Process & Operations

About the Cluster

This cluster explores the internal workings of the laundry process which includes the operational steps taken from the moment clothes are submitted to the time they are returned. It includes human effort, logistical mechanisms, and task efficiency. Every action like logging, tagging, washing, drying, and delivery is part of a tightly interlinked chain, and disruptions or inaccuracies in any one part can lead to major breakdowns across the system. The key insight here is that while the system is meant to serve a high volume of users, it is often manually run and lacks standardization, causing inconsistencies, delays, and errors.



Logging Time per Student

The amount of time staff spend on recording each student's laundry submission. Longer logging times can lead to queue build-ups and slower processing overall.

Tagging Accuracy

The reliability of the tagging process in correctly identifying which clothes belong to whom. Errors here are a major cause of loss and confusion.

Daily Laundry Cycles

The number of times the laundry process is completed in a day, from washing to drying and returning. Fewer cycles may indicate bottlenecks or insufficient staff.

Staff Count

The number of personnel assigned to the laundry process. Fewer staff result in increased workload, lower accuracy, and greater chances of error.

Tagging Time

The time it takes to attach identifying tags to clothes. This impacts how quickly garments move through the pipeline and whether they can be tracked effectively.

Tagging Method

The approach used to mark or tag clothes (e.g., manual paper tags, safety pins, color codes). Some methods are more prone to falling off, getting damaged, or being misread.

Delivery Time

The average time taken to return clothes to students after submission. Delays contribute directly to dissatisfaction and system overload due to backlog.

Clothes Load per Day

The total volume of clothes processed daily. A mismatch between load and capacity is a direct contributor to system strain and inefficiency.

Logging Accuracy

The correctness and consistency of recorded student information. Inaccurate logging can result in misplaced or unclaimed clothes.

Garment Damage Rate

The frequency with which clothes are returned torn, faded, or otherwise damaged. This not only reduces trust in the system but also increases staff-student conflicts.

Lost Clothes Rate

The number of garments lost or unaccounted for in each cycle. This is a critical metric of system failure and deeply impacts trust and fairness.

System Design & Infrastructure

About the Cluster

This cluster deals with the physical and structural components of the laundry system of how it is built, how resources are allocated, and how reliably the system can perform under regular and high loads. The system design directly influences how efficiently the processes run and how much stress the staff and users experience during peak demand.

A poorly designed infrastructure not only slows down operations but also creates friction points that amplify other issues, such as errors, delays, and misuse.



Submission System Type

Refers to whether the process of submitting clothes is manual, token-based, digital, or unstructured. The submission method determines how orderly and predictable the laundry intake is, affecting all subsequent steps.

Machine Quality

Refers to the condition and efficiency of washing and drying machines. Older or poorly maintained machines can damage clothes, break down often, and increase cycle times significantly..

Tag/Bag Availability

The consistency and sufficiency of resources like tags or laundry bags. Shortage or poor quality of these can delay submissions, increase confusion, and raise the chances of clothes being mixed up.

Schedule Reliability

How consistently the laundry process follows a fixed schedule for both for pickups and returns. Unreliable schedules create frustration among students and reduce accountability within the system.

Laundry Room Capacity

The amount of physical space available to process laundry including sorting areas, storage, washing machines, and drying space. Overcrowded or poorly arranged rooms create bottlenecks and stress for staff.

Machine Maintenance rate

The frequency and quality of machine servicing. Low maintenance leads to frequent breakdowns, while a proactive schedule can ensure smoother operations and reduce delays or damages.

Communication and Feedback

About the Cluster

This cluster focuses on the exchange of information between the system and its stakeholders including students, staff, and administration. A system that lacks clarity in rules, weak reporting mechanisms, and poor feedback loops tends to breed confusion, frustration, and mistrust. Communication is not just about delivering instructions – it's about creating transparency, ensuring people feel heard, and enabling continuous improvement through responsive feedback. Failures in this cluster often lead to misinformation, unresolved issues, and the inability to detect patterns of failure or dissatisfaction.



Rule Clarity

Refers to how clearly the submission rules, garment limits, timelines, and procedures are communicated to students. Ambiguity in rules often leads to accidental violations or unchecked abuse of the system.

Student Impact Awareness

Measures how aware students are of how their actions (such as violating garment limits or submitting proxies) impact the system and the staff. Low awareness leads to low accountability and misuse of the service.

Staff Reporting Channels

The formal or informal mechanisms that allow laundry staff to report problems, raise concerns, or suggest improvements. When these channels are missing or ineffective, operational issues go

Status Update Transparency

Refers to how easily students can check the status of their laundry, whether it's been washed, dried, delayed, or returned. Lack of visibility results in anxiety, repeated questioning, and overcrowding at pickup counters.

Feedback Loop Strength

Indicates the robustness of the system's ability to gather input from students and staff and implement changes based on it. Strong feedback loops help in resolving recurring issues and signal responsiveness from administration.

System Performance Metrics

About the Cluster

This cluster focuses on the measurable outcomes and health indicators of the overall laundry system. These variables capture how the system is functioning in practice – from service speed and reliability to stakeholder satisfaction and friction points. By monitoring these metrics, we can evaluate the success or failure of process changes and identify whether improvements are genuinely benefiting users and staff.

This cluster is crucial because it acts like a diagnostic dashboard. While other clusters represent underlying causes and behaviors, this one helps us understand symptoms and track progress over time.



Average Delivery Time

The time taken from the submission of clothes to their return. It's a core service metric reflecting system efficiency and consistency.

Complaint Rate

Measures the number of complaints raised by students or staff. A high rate points to dissatisfaction, but also reflects the presence (or absence) of proper feedback mechanisms.

Delay Frequency

How often delays occur in the laundry process. High delay frequency signals recurring bottlenecks, either from overloaded capacity, poor scheduling, or equipment issues.

System Efficiency

System Efficiency
A more holistic measure capturing how well resources (time, people, machines) are being utilized to deliver timely, accurate laundry service. It helps track the impact of any intervention.

Staff Satisfaction

Indicates how laundry staff feel about their workload, processes, support systems, and recognition. Low satisfaction often correlates with inefficiency, mistakes, or high turnover.

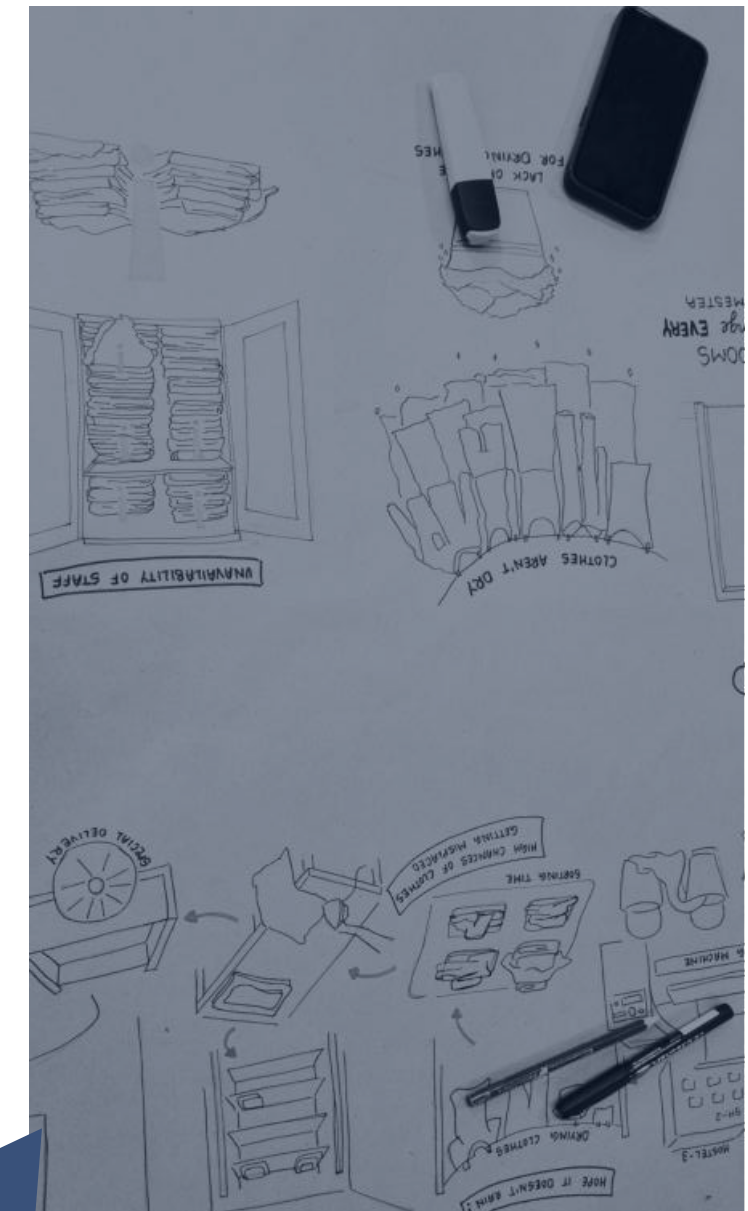
System Mapping and Problem Visualization

Objective

The objective of this stage was to construct a comprehensive and accurate visualization of the current laundry system, with a focus on identifying existing inefficiencies and problem areas. Rather than proposing solutions, this phase was dedicated to understanding the system's structure, patterns, and pain points.

Sequential System Visualization

A visual walkthrough of the laundry process was created using sketches and on-site photographs. This walkthrough captured every step – from the point a student submits their clothes, through sorting, tagging, and washing, to the final return of the laundry. Each step was annotated with observations and indicators of operational friction.



These sketches were not only illustrative but analytical – highlighting the flows, blocks, and loops that contribute to systemic overload.



Influence Map: Effects and Causes

Parallel to the visual system representation, an influence map was constructed to establish causal relationships between observed effects and their root causes. This map was used to trace the chain of impact from deep system variables to surface-level issues.

Each of the following effect was traced back to multiple systemic causes:

- Key Effects Observed
- Delays in pickup and delivery
- Mix-ups and misplaced clothes
- Poor communication between students and vendors



Effect 1 : Delays in pickup and delivery

The time taken from the submission of clothes to their return. It's a core service metric reflecting system efficiency and consistency.



Effect 2 : Poor communication between students and vendors

How often delays occur in the laundry process. High delay frequency signals recurring bottlenecks, either from overloaded capacity, poor scheduling, or equipment issues.



Effect 3 : Mix-ups and Misplaced Clothes

Indicates how laundry staff feel about their workload, processes, support systems, and recognition. Low satisfaction often correlates with inefficiency, mistakes, or high turnover.

Sites of Intervention

Through this map, the following key intervention points were identified:



The manual logbook used for tracking clothes.



The lack of communication or reminders for uncollected garments.



The absence of structured student-vendor communication.



The eight-clothes submission rule and its enforcement.



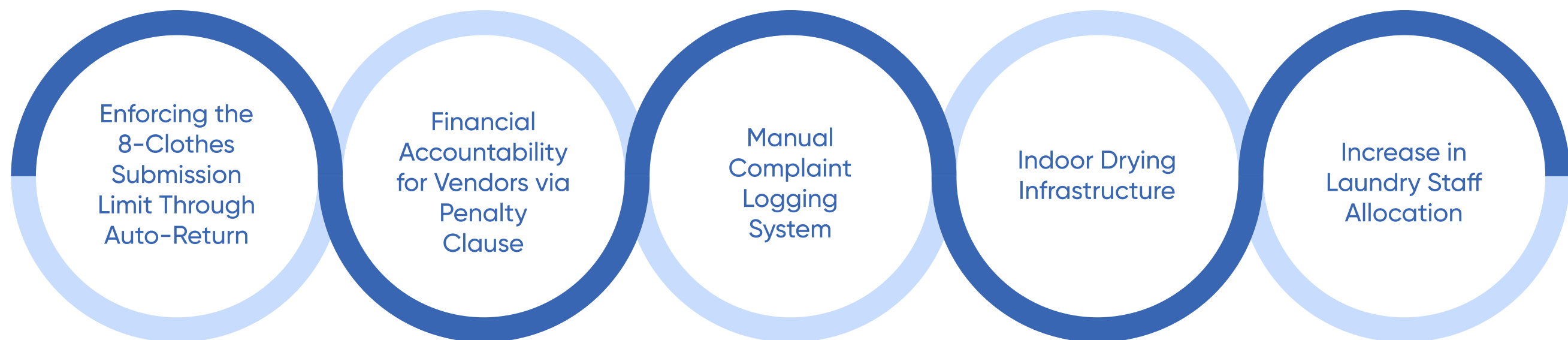
The manual tagging system, based on room numbers.

These were later used to define leverage points and potential redesign strategies in subsequent stages.

Leverage Points

In systemic design, leverage points refer to strategic areas within a system where small, focused interventions can produce significant and lasting change. Based on stakeholder research, process mapping, and cause-effect analysis, five such leverage points were identified within the college laundry system.

These were chosen based on their potential to disrupt negative feedback loops, reduce system load, and create measurable improvements in fairness, efficiency, and accountability.



Untangling the Knots

Understanding Overload in Our College Laundry System

The college laundry system is strained by overload, where rule violations, manual operations, and infrastructure gaps create a cycle of delays, garment loss, and staff burnout. As trust erodes and fairness is questioned, bypass rules, further stressing the system. This tangled web of behaviors and inefficiencies highlights the need to understand deeper causes before proposing meaningful, lasting improvements.

Gigamap

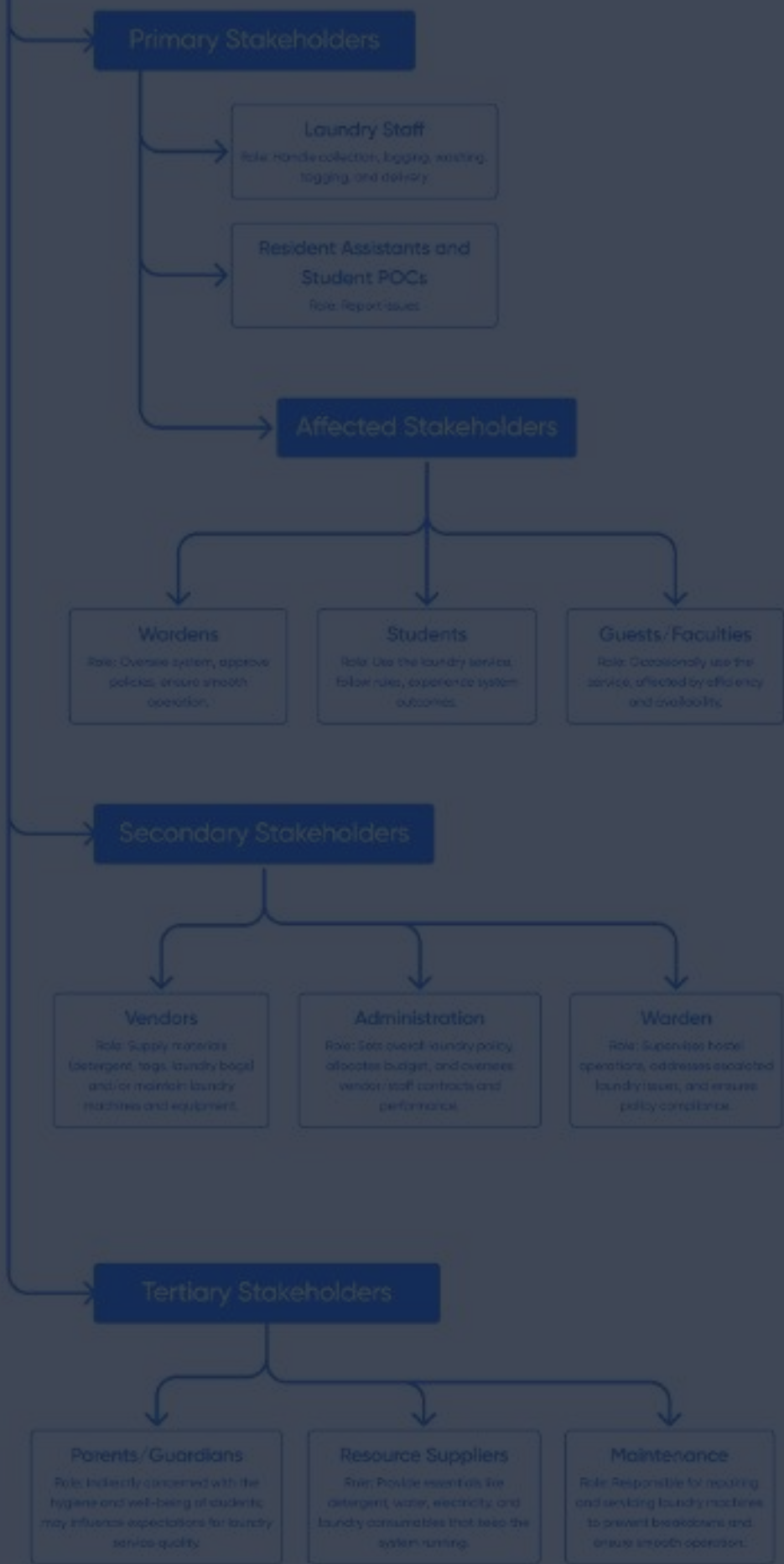
Untangling the Knots

The Giga Map serves as a comprehensive visual synthesis of the hostel laundry system, capturing its complexity through a single interconnected diagram. It consolidates the identified problem areas, stakeholders, leverage points, sites of intervention, and the system map into one cohesive representation.

The chosen title, Untangling the Knots, reflects the process of untangling intricate relationships, dependencies, and challenges within the system, transforming them into structured and actionable insights.

There is no proper way to talk to the vendors. The system to complain is too complicated and slow.

Stakeholders



Concept and Metaphor

The Giga Map combines the form of a tree with the metaphor of threads to depict the interconnected elements of the laundry ecosystem.

1. Roots

Core variables causing inefficiencies and delays.

2. Branches

Variables that are not root causes but still shape and affect the problem, forming part of the process.

3. Trunk

The central flow linking causes to other influencing variables.

Purpose

This map layers causes, processes, and interventions into one framework, allowing stakeholders to:

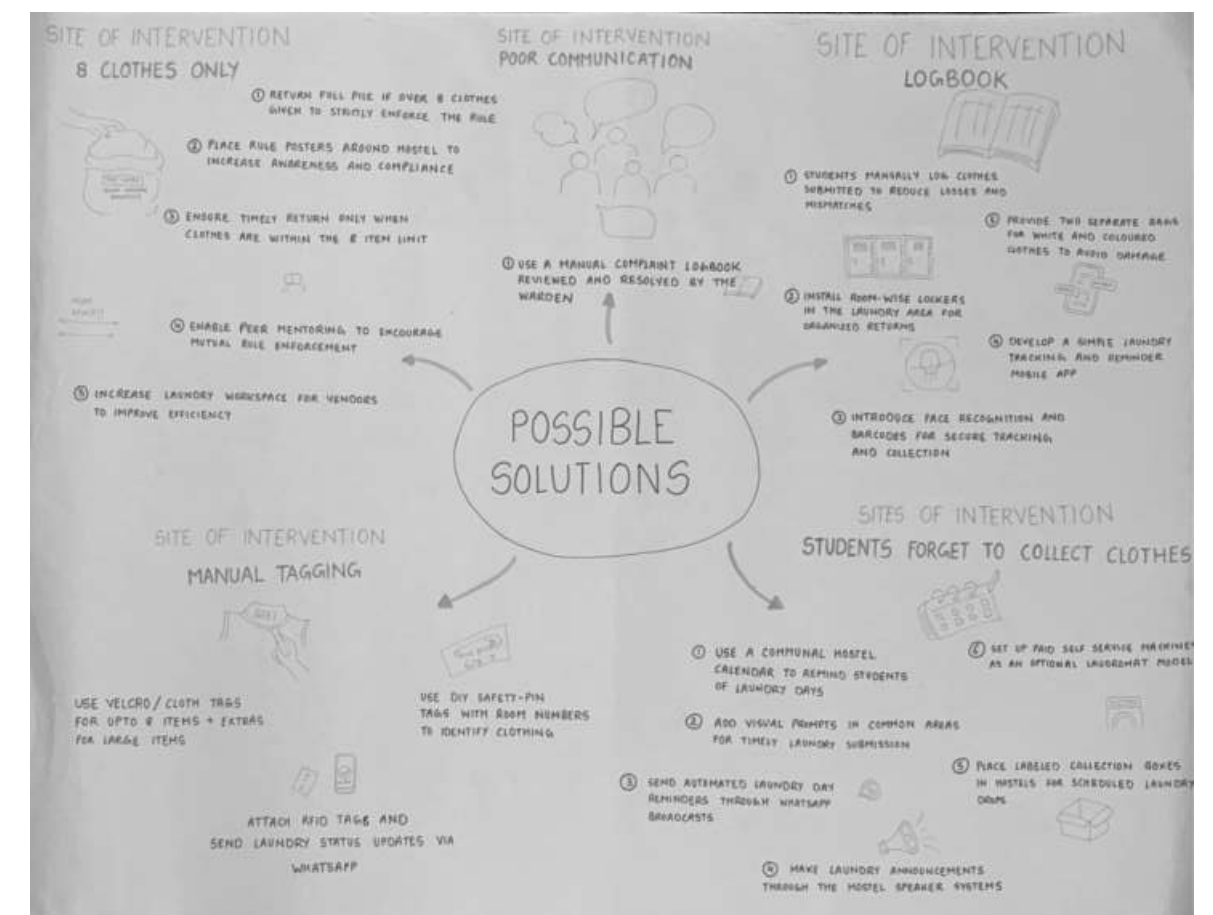
- See the system's full complexity.
- Identify leverage points for high-impact change.
- Trace relationships from root issues to influencing factors.
- Recognise interdependencies across roles and operations.

Proposed Interventions

Idea Pool

To address the inefficiencies and systemic failures identified within the college laundry system, a wide range of potential interventions were ideated.

These solutions directly correspond to the previously identified five key sites of intervention and were developed to improve accountability, reduce system load, enhance communication, and introduce technological or process efficiencies. Each proposed intervention is contextualized below.



Full Pile Returned if Over Limit

To ensure strict adherence to the 8-item limit, the system would automatically return all clothes if more than eight are submitted. This measure introduces a strong deterrent to rule violation and reinforces submission discipline.

Peer Monitoring Model

Encouraging students to self-regulate through informal peer monitoring strengthens community accountability. Mutual reinforcement of the rules is expected to increase collective compliance while reducing dependence on staff enforcement.

Rule Awareness through Visual Signage

Posters and visual indicators placed across hostels serve as continuous reminders of the laundry rules. This increases awareness and visibility of submission constraints, especially for new students or those unaware of existing policies.

Expand Vendor Workspaces

Increasing the physical space allocated to laundry vendors allows for better sorting, tagging, and handling of clothes. This infrastructural change aims to reduce errors and improve efficiency under the existing workload.

Timely Delivery Conditional on Rule Adherence

Clothes will be returned on time only if the submission adheres to the 8-clothes rule. Students exceeding the limit will face delivery delays, thereby creating a natural consequence for over-submission.

Timely Delivery Conditional on Rule Adherence

Clothes will be returned on time only if the submission adheres to the 8-clothes rule. Students exceeding the limit will face delivery delays, thereby creating a natural consequence for over-submission.

Manual Complaint Logbook

A physical complaint register, regularly reviewed by the hostel warden, enables students to document grievances such as delayed or lost clothes. This solution enhances transparency and establishes a formal grievance redressal mechanism.

Room - Wise Lockers for Laundry Returns

Dedicated lockers based on room numbers streamline the process of returning clothes and ensure that laundry is organized systematically. This reduces confusion during distribution and supports privacy and order.

Student -Led Manual Logging

Students personally recording the number and type of clothes submitted minimizes errors and disputes in case of loss or misplacement. This intervention introduces a check-and-balance model within the submission process.

Barcoded and Biometric Tracking

Face recognition systems and barcoded tags can automate the tracking of laundry submissions and returns. This technology-driven solution ensures precision and allows students to track their clothes digitally.

Segregated Laundry Bags

Providing students with two separate bags, one for white and one for colored clothing, helps reduce garment damage during collective washing. This simple sorting mechanism addresses one of the common complaints among students.

Mobile Laundry App

A lightweight mobile application that notifies students about laundry pickup status, expected delivery times, and missing clothes introduces traceability into the process and bridges the communication gap between staff and students.

Velcro Cloth Tags

Reusable Velcro tags marked with room numbers can be attached to garments, particularly for the allotted 8 clothes. This intervention is cost-effective, easy to implement, and sustainable in the long term.

Hostel Calendar For Laundry Days

A shared hostel calendar clearly marking laundry days acts as a visual cue for students, minimizing missed submissions and late pickups.

RFID Integration with WhatsApp Updates

RFID-enabled tags attached to clothes, paired with automated WhatsApp updates, allow students to track the status of their laundry in real time. This semi-automated system improves both visibility and traceability.

Visual Prompts in Common Areas

Strategically placed reminders in high-footfall areas of the hostel prompt students to prepare their laundry on time. These visual nudges help bridge the gap between intent and action.

Safety-Pin Room Number Tags

An affordable and DIY alternative, these tags use safety pins and fabric-safe labels to tag clothes with room numbers. While simple, they significantly reduce chances of mix-ups and are easy to implement.

WhatsApp reminder Broadcasts

Automated broadcast messages remind students of laundry collection timings. This low-effort digital solution is well-aligned with students' communication habits and ensures timely action.

Shortlisted Solutions

From the full set of 20 proposed interventions, two solutions were selected for prototyping and implementation based on feasibility, expected impact, cost-effectiveness, and alignment with key leverage points in the system:

1. RFID-Enabled Cloth Tagging System



This solution integrates RFID tags into the laundry workflow to automate tracking, minimize errors, and introduce digital accountability. Each student receives a set of waterproof, heat-resistant tags to be attached to their garments. These are scanned at both the point of submission and return, creating a traceable loop that prevents loss or misplacement. The system enforces a hard limit of 8 items per wash and flags discrepancies instantly.

2. Visual Clues and Reminders for Compliance



Complementary to the RFID system, strategically placed visual prompts (e.g., rule posters, illustrated instructions, and calendar boards) are used to reinforce behavioral compliance. These are placed in hostels, laundry rooms, and common areas to guide students through submission rules, timelines, and best practices, thereby increasing awareness and accountability.

Implementation and Testing

The implementation and testing phase of the proposed RFID-based laundry tracking system was conducted. The objective was to simulate the intended system's functionality, test its feasibility, and receive feedback from the primary service provider – the laundry vendor.

Prototype Development – To simulate the tracking process, we began by building a basic RFID scanner setup using Arduino Uno. Once the scanning functionality was verified, we transitioned to ESP32, which enabled wireless connectivity



Stakeholder Interaction and Feedback

After the prototype was functional, it was presented to the current laundry service provider. He was walked through the scanning, tagging, and logging process using the live Firebase interface....

His response was notably positive. The following feedback was documented from the laundry vendor:

Ease of Use

The vendor found the RFID scan-and-log system simple, intuitive, and faster than the existing handwritten method.

Time Saving

The new system eliminates the need for writing room numbers and lists in logbooks, significantly reducing intake time per student.

Error Reduction

Automated linking of tags to student IDs prevents mismatches and misplaced clothes.

Lost Clothes Prevention

By assigning a tag to each garment, the system enables complete traceability from submission to delivery.

Tag Durability

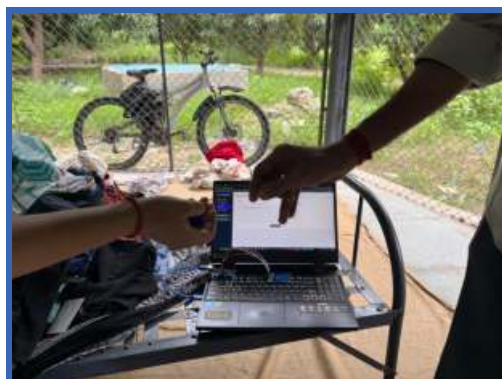
Feedback confirmed that a removable and reusable tag design was preferred for ease of use and maintenance.

Vendor Interest

The laundry contractor confirmed willingness to invest if the system is formally implemented by the hostel management.

Scalability Potential

The model has potential to be extended to other hostels or educational institutions with similar laundry service structures.



Initial prototype setup using Arduino and RFID module at laundry site.



Hands-on trial by the laundry manager - testing the RFID scan and feedback on its usability.



Explaining how the RFID tags would be attached to the laundry bags and clothes.



Explaining the system interface and tag functionality to the laundry manager.



Discussion with the main laundry vendor regarding the feasibility and interest in implementing the system.

RFID Tag Design

A key outcome of this phase was the development of a reusable, durable RFID tag design tailored to the needs of students and th laundry staff.

Tag Specifications

- Quantity per Student: 16 reusable RFID tags (8 per laundry bag)
- Material: Made from waterproof, heat-resistant silicone
- Attachment Mechanism: Snap-press closure (similar to badge clips or baby bib fasteners)
- No loops, stitching, or permanent fixtures required

This design enables:

- Easy tagging of garments without damaging fabric
- Quick removal before ironing or drying
- Seamless reuse throughout the semester



Addressing Common Concerns

During the class presentation and peer review, several pertinent questions were raised by fellow students, hostel residents, and faculty. These are documented below, along with the corresponding system responses:

What if students buy more than 12–16 tags?

The system has an inbuilt limit-check. If more than 8 garments are submitted per cycle, the system flags and rejects the load.

Isn't this too much hassle for students?

Compared to other hostel systems that require manual logging, this simplifies the process while introducing accountability.

Why not use store-bought RFID tags?

Retail tags are not waterproof or heat-resistant. Our custom tags use durable materials like silicone or polycarbonate.

Why limit students to 12–16 tags?

Limits align with machine load capacity and hostel policy. Regular washes allow 8 clothes, special days (15th & 30th) allow 12–16.

Can one tag be used for multiple clothes?

Each tag is mapped to one item only. If students bundle items, they assume responsibility for losses.

Why not use stickers?

Stickers peel off during washing, melt during ironing, and can damage clothing. Silicone tags are designed to avoid such damage.

What will it cost?

Approx. ₹15–₹30 per tag, totaling ₹200–₹300 per student per semester. Tags are reusable and require no replacement unless lost.

Team Members



Abhineet Kumar



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